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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	CodeCatalyst
Selected Domain	Domain 1 - AIVERSECITY
Selected SDG Track (1–17)	SDG4、SDG8
Project / Solution Name	CareerBridge AI
Name (Leader)	FANG ZIMING
Email & Phone (Leader)	fangziming074@gmail.com +601125963708
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Rapid AI advancement is reshaping the labour market, automating numerous routine and knowledge-intensive jobs. Many workers face displacement and urgent reskilling demands, while university graduates also encounter tough employment challenges, as their skills often fail to keep up with evolving job requirements.

This structural labour crisis poses serious economic and social risks. Mass unemployment undermines inclusive economic growth and contradicts the pursuit of decent work. It may also widen social inequality: workers and young graduates without proper guidance are easily marginalised amid the AI transition. Existing career resources remain scattered and one-size-fits-all, lacking predictive insights and personalised transition plans.

Our system forecasts rising and declining job roles and delivers customised upskilling roadmaps. It helps workers, career transitioners and students adapt to labour market shifts, build future-ready competencies, and access sustainable quality jobs, aligning with SDG 4 and SDG 8.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

CareerBridge AI is a fully deployed web application built for students and career transitioners navigating AI-driven labour market shifts, aligned with SDG 4 and SDG 8. We prototyped and orchestrated the overall system architecture on Replit, the official environment provided in SDC workshops. Users select their target region, occupation and time horizon, input existing skills, and obtain cited, forward-looking career forecasts alongside personalised skill-gap learning pathways. The platform calculates a readiness score reflecting skill coverage for target roles, to guide upskilling rather than guarantee employment outcomes.

Two distinct AI pipelines power the system. First, it leverages DeepSeek Search and DeepSeek-V4-Pro to retrieve and synthesise up-to-date labour sources with citations, generating regional career outlooks (e.g. 1–

3 year forecasts for Product Managers in Malaysia). Second, the authenticated AI workspace sends user profile and career context to a RunPod Serverless endpoint running Qwen3-0.6B, which generates custom learning plans, tutoring content and mock interview feedback.

The frontend uses HTML, CSS and JavaScript. The Node.js backend stores user profiles in PostgreSQL and is hosted on Render. It also integrates World Bank WDI unemployment datasets for macroeconomic context. This implementation applies Replit architecture orchestration, GenAI APIs, relational database design and cloud deployment skills covered in SDC Phase 1 workshops. Live demo: https://codecatalyst-careerbridge-ai.onrender.com/ GitHub: https://github.com/fangziming074-creator/codecatalyst-careerbridge_ai

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

We build this functional deployed prototype with a full stack orchestrated and prototyped on Replit, the official tool provided by SDC organisers.

Browser (HTML/CSS/JavaScript) → Render Node.js 20 web service → PostgreSQL via pg (for user accounts, profiles and saved CareerBridge states).

Two core AI workflows run on top of this stack:

1. Market-report pipeline: Node.js → DeepSeek Search (5 targeted searches & source retrieval) → DeepSeek-V4-Pro → cited labour outlook report → browser.

2. Authenticated AI workspace pipeline: browser → Node.js → RunPod Serverless OpenAI-compatible endpoint (Qwen3-0.6B) → outputs custom learning plans, tutoring content and mock-interview feedback.

RunPod Serverless is our integrated sponsor cloud platform, delivering model inference for the logged-in AI workspace. DeepSeek powers real-time source retrieval and market report synthesis.

All sensitive credentials, including `DEEPSEEK_API_KEY`, `RUNPOD_API_KEY`, RunPod endpoint configurations and `DATABASE_URL`, are stored as server-side environment variables on Render. They are never exposed on the frontend or committed to the public repository.

This architecture unifies the workshop components: Replit-based architecture prototyping, GenAI APIs, relational database storage and cloud hosting.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team adopted design thinking to build a technically feasible, intuitive solution deployable for Phase 2. We prioritised a core user need: students and career transitioners require transparent, evidence-backed labour-market trend analysis and actionable upskilling roadmaps. This guided our workflow: select region, occupation and timeline, review cited forward-looking career outlooks, self-audit skills against role requirements, and follow personalised learning recommendations.

Our live prototype distinguishes user self-reported data from AI outputs, attaches source citations for all trend forecasts, and clarifies that the readiness score serves only for planning rather than predicting hiring outcomes. DeepSeek powers the critical trend research and synthesis pipeline, forming the backbone of our evidence-based career insights, while RunPod delivers the learning assistant.

Formal user research is yet to be completed. For Phase 2, we plan 5–8 usability interviews and a 20-participant Malaysia pilot. Metrics include task success rate, source comprehension, 14-day learning uptake and career-planning clarity scores. Feedback will strengthen trend coverage, onboarding, privacy notices and learning-path usability.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	UN SDG 4, Target 4.4: By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship. UN SDG 8, Target 8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value. CareerBridge supports informed skills planning and transparent labour-market trend analysis for students and career transitioners; it does not directly create jobs. Sources: https://sdgs.un.org/goals/goal4 and https://sdgs.un.org/goals/goal8
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Global youth unemployment stood at 12.4% in 2025, representing 67 million unemployed people aged 15–24 (students and new graduates). Separately, ILO estimates that 16.2% of the global workforce (553 million workers) hold roles exposed to substantial AI-driven task transformation, including mid-career workers facing urgent displacement and career transition pressure. Sources: ILO Global Employment Trends for Youth 2026; ILO Working Paper on Generative AI and Jobs. These global benchmarks frame the broader labour challenge; CareerBridge's pilot-specific baseline has not yet been collected. Before launching the pilot, we will measure participant task

Metric Category	Your Response
	completion rates and career-planning clarity scores to establish our own product baseline for comparison.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Primary KPI: percentage of pilot participants who, within 14 days, produce a source-cited labour market outlook report, finish a role-specific skill-gap assessment, and record at least one learning action. Secondary KPI: paired change in self-reported career-planning clarity on a 1–5 scale. Baseline data will be captured prior to first use; final reporting will include numerator, denominator and survey attrition rate.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	By the final Phase 2 prototype, run a pilot with at least 20 participants, including both students/recent graduates and mid-career workers facing career transition pressure. Target: at least 70% (14 out of 20) complete the source-cited labour market outlook report → role-specific skill-gap assessment → first learning action sequence within 14 days; the mean paired career-planning clarity score rises by at least 1 point on a 1–5 scale. This is a pilot behavioural target, not a claim of job placement or reduced unemployment.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

TEAM CONFIRMATION REQUIRED: an earlier CareerBridge AI version existed before this rework. Confirm whether any code, design or research predates the hackathon start, what is reused, and which member(s) own it; then select Yes or No before signing.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Node.js 20 runtime and built-in modules	Runtime and built-in code	Node.js MIT; bundled components retain their own licenses.
pg npm package and transitive dependencies	Open-source database driver and dependencies	MIT for pg and most locked packages; ISC for pg-int8 and split2. See package-lock.json.
DeepSeek Search API and DeepSeek-V4-Pro	Hosted AI search/model service	DeepSeek API/service terms; proprietary hosted service; model weights are not in the repository.
Qwen3-0.6B served	Pre-trained model	Apache-2.0 model license;

through RunPod Serverless		official model card: https://huggingface.co/Qwen/Qwen3-0.6B. Weights are not in the repository.
RunPod Serverless endpoint	Cloud inference service	RunPod service terms; hosted infrastructure.
Dynamic web content retrieved via DeepSeek Search	Dynamically fetched third-party source data	Terms defined by individual webpage publishers. Generated reports include direct source links for attribution.
World Bank Data API, indicator SL.UEM.TOTL.ZS	Public labor-market data API	World Bank dataset terms (default CC BY 4.0 unless indicator metadata specifies otherwise); attribute the indicator source as listed in metadata.
OpenDOSM Monthly Youth Unemployment dataset (proposal baseline)	Public statistical dataset	CC BY 4.0; credit Department of Statistics Malaysia (DOSM). https://open.dosm.gov.my/data-catalogue/lfs_month_youth?visual=u_rate_15_24

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

TEAM CONFIRMATION REQUIRED: ask all five members whether an employment, internship or consulting agreement could claim this work. Select Yes or No; if Yes, identify the member and describe the relevant restriction before signing.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] Fang Ziming

[Date] 2026.10.24